Cloudberries

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Data scientist

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Rafael is an accomplished developer and analyst with a passion for Data Science. He has a PhD (Ph.D) in modeling, simulation and analysis at NTNU, where he studied how to optimize chemical processes and make them more environmentally friendly.

He has over 10 years of experience in developing, operating and managing IT solutions on various technologies and platforms such as .Net, Python, Azure and GCP. He also has experience in facilitating solutions from "on-premises" to the cloud.

Recently, Rafael has worked as both a Data Engineer, Data Scientist and Back-end Developer in various projects at Elvia. One of these, for example, was about creating and commissioning image recognition models that help the customer detect anomalies faster.

Rafael is solution-oriented, easy to work with and is committed to delivering high-quality solutions. Rafael thrives as part of a larger team and with pair programming. Rafael enjoys complex problem solving and easily familiarizes himself with new tools and frameworks.

Project experience

DURATION	PROJECT
09.2024 - 02.2026	**Sparebank 1 Insurance: PS27** The PS27 project deals with a replacement of the existing core system F2100 with a new system. The core system F2100 is supplied by the Danish company FDC. The project's goal is to start operations in 2027. **ROLLE: Data Scientist** In this role, Rafael explored the landscape for the generation of synthetic data, with a particular focus on privacy. He studied various synthetic data generation tools and their underlying model architectures, prioritizing flexibility, resiliency, and scalability due to tight deadlines and uncertain requirements. Differential privacy was identified as an important mathematical framework, as

it offers quantifiable guarantees related to the risk of re-identification, and explicitly addresses the inherent trade-off between the strength of the privacy safeguards and the realism of the synthetic data generated.

Rafael recognized the complexity of generating synthetic versions of relational databases, a known challenge in the field. After a thorough assessment of the available solutions, he identified a method described in a research article as the most suitable. This method employed a differentially private multi-table synthesis algorithm, designed to link independently generated synthetic tables from single-table synthesis tools, by linking tables in pairs, while preserving statistical patterns and data distributions across tables.

The chosen algorithm used marginal queries across tables and minimized a loss function to ensure good quality in relation to sharp data. Crucially, this approach enabled a modular approach, breaking down the complex problem of synthetic database generation into manageable steps.

To ensure that all generated synthetic data was of sufficient quality, and that the privacy of the training data was actually safeguarded, Rafael devised a set of metrics that are calculated automatically upon synthesis.
The quality of the synthetic data was measured using metrics such as fidelity, utility and integrity. These quantified, among other things, how well the data distributions in the real data were recreated in the synthetic data.
Privacy metrics such as Distance to Closest Record and Membership Inference Attack measured how well the privacy framework – differential privacy – protected the training data.

ROLE: Advisor

In this role, Rafael was responsible for the planning and execution of the synthetic data initiative, in close collaboration with relevant stakeholders. Rafael served as a special resource responsible for getting deep into both the GDPR regulations and the organization's needs, and then developing a strategy that took both into account. This involved close collaboration with legal experts and other professional communities to conduct internal risk assessments.
The work also included knowledge sharing related to privacy principles, as well as presenting results and recommendations to different teams in the organization.

ROLE: Python Developer

In his role as a Python developer, Rafael created RelationalSynth, a Python library designed to automate differential private multi-table synthetic data generation. This library utilizes SmartNoise, an open-source differential privacy library that offers single-table synthesizers to generate individual synthetic tables. RelationalSynth then integrates a differentially private algorithm from a 2024 research paper to connect these tables. The multi-table algorithm connects synthetic tables in pairs, handles referential integrity, and preserves relationships between parent-child tables. The core of the algorithm is to iteratively solve a large system of linear equations. To ensure differential

privacy, noise injection was strategically implemented where sharp data was accessed. Rafael developed this algorithm with PyTorch, focusing on optimizing for sparsity and vectorization to improve performance. RelationalSynth was built to reuse as much as possible of SmartNoise's functionality for preprocessing data. The library's code structure was heavily inspired by SmartNoise, mimicking the design, with adaptations where necessary due to the differences in the issues addressed. To handle the computationally intensive nature of the multi-table algorithm, a dedicated server equipped with a GPU card was set up. Rafael collaborated with several teams, including the "Information and Insights" team, to install the GPU and navigate dependency challenges to ensure compatibility between Torch and CUDA versions. RelationalSynth simplifies the process to specify a set of SQL queries that define the sharp database tables, along with some supporting metadata and dependency information. Upon successful generation, RelationalSynth seamlessly inserts the synthetic data into a SQL Server database.

ROLE: Fullstack Developer

Rafael developed a prototype for a microservices application to realize the vision of synthetic data. The goal of the app was to democratize the generation and use of synthetic data by providing a user-friendly and self-service solution, where anyone can create synthetic versions of real databases as needed.

Since the customer has an extensive AWS Kubernetes ecosystem, the application was architected to run in the cloud. The application consisted of a FastAPI backend, a Vue.js frontend, and a PostgreSQL database for application data. The architecture also included file storage for both temporary storage needs and for permanent storage of the generated synthetic results, Python worker nodes with support for both CPU and GPU, as well as Redis for job queues and real-time progress updates via pub/sub. Each component was mapped to corresponding AWS services, such as Amazon ECS for backend and worker nodes, S3 for storage, and RDS for the database.

The prototype was successfully demonstrated to relevant stakeholders locally using Docker Compose. The further rollout to AWS was put on hold due to strategic changes and uncertainty about the project's future direction.

EXPERTISE

SQL, Python, Personvern, GDPR, PyTorch, NumPy, Pandas, Forsikring, Research, MS SQL Server, Kubernetes, Vue.js, FastAPI, Redis, JavaScript, Docker Compose, AWS

03.2025 - 08.2025

Sparebank 1 Development: The anti-fraud service of the future

The aim of the project is to replace the existing internal anti-fraud system ASLA with a new and more comprehensive system. This involves a procurement process to purchase a new solution for the SpareBank 1 alliance from one of the leading players in the sector. At the same time as the procurement process is ongoing, the project also includes important work to increase the degree of fraud detection.

ROLE: Data analyst

Rafael participated in a high-priority project with the goal of replacing the customer's existing internal anti-fraud system (ASLA) with a new and modern system. Under tight deadlines and in close collaboration with Arne Kristian (lead architect/data scientist) and Jostein (system administrator and creator of ASLA), Rafael helped build a bridging solution that also served as a critical toolbox to extract insights from raw and previously unused data.

The project started with a deep exploration phase where Rafael combined logs with data from fraud cases to distinguish positive and negative transactions. This analysis revealed valuable demographic insights, such as that older women were most vulnerable as victims, while young men were overrepresented among the fraudsters.

A main challenge was that it was difficult to find effective rules without better profiling. To solve this, Rafael created a custom "sandbox" for building profiles, i.e. historical snapshots of relevant entities (an entity in this context can be e.g. a customer or a counterparty's account). Manual rule identification proved to be time-consuming. To uncover whether it was even possible to identify fraud with the available data, Rafael turned his focus to training advanced classification models (Random Forest, Gradient Boosting) on profile attributes. These models proved that it was actually possible to detect about 4% of all fraud cases with high precision.

To operationalize the new rules in the existing system, Rafael adapted an open-source solution compatible with Scikit-learn-based ensemble models. This allowed the team to "reverse engineer" the ML models into interpretable rules. This approach paid immediate value, and the foundation Rafael helped build was later used by the team to roll out a full-fledged machine learning solution in production.

ROLE: Database Developer

Responsible for enabling rapid hypothesis testing, Rafael architected a custom sandbox solution built entirely in PostgreSQL. Through a close collaboration with the system's creator, Jostein, Rafael ensured that the solution was well integrated with the existing architecture. The sandbox consisted of three layers: a profiling layer for building profiles, a rules engine, and an experiment administrator (orchestrator).

A key requirement was the ability to easily test new profile attributes, rules, and entities. Raphael made this possible by generalizing the core logic and defining standardized function and procedural signatures. This architectural choice served as a consistent interface. Rafael often made use of PostgreSQL's JSONB objects, which provided the necessary flexibility in the data schema without the need for rigid structural changes.

Performance was critical. Rafael leveraged custom indexes and refactored the procedure code to minimize runtime. By vectorizing functions and procedures,

he achieved speed improvements of several orders of magnitude. This optimization allowed the system to generate millions of profiles—snapshots of an entity's state at any given time—quickly enough for production use. In his final weeks, Rafael worked with two new in-house developers to roll out these optimized procedures.

ROLLE: Data Engineer

Rafael worked alongside a team of consultants and in-house staff to transform raw, unused audit logs and data from known fraud cases into structured data suitable for fraud detection.

A critical technical challenge was to build the profiles in the sandbox. This layer had to generate historical snapshots that captured the state of an entity (e.g., an account's average transaction amount or known counterparties) at any point in time in the past.

Rafael developed this solution directly where the data was located to avoid latency. The sandbox transformed audit and fraud data to create robust datasets for training and testing. The architecture relied heavily on PostgreSQL functionality; Rafael used JSONB for data flexibility and implemented custom indexes to ensure queries ran efficiently.

The solution was designed for rapid iteration. By making sure the logic was generic, Rafael made it easy to extend the pipeline with new profile attributes, rules, and entities without breaking existing workflows. In the final phase of the project, Rafael focused on the operationalization of the solution, working hands-on with two new employees to deploy the profiling logic to the production environment. This laid the foundation for the client's future ML-based anti-fraud capabilities.

ROLE: Solution Architect

In collaboration with the Chief Architect, Arne Kristian, Rafael designed a flexible framework to meet the client's urgent needs for insight, detection and fraud reduction. Rafael provided the project with a toolbox that could work alongside the legacy ASLA system while paving the way for future modernization.

One of Rafael's most important contributions was the design of a three-layer sandbox (profiling, rules, experiments) that decoupled the various steps in the process. Rafael designed the logic of the sandbox to be very generic. This abstraction allowed the team to easily configure new entities, profile attributes, and rules without having to rewrite the underlying code.

Rafael was also instrumental in the decision to keep the heaviest processing jobs in the database layer, leveraging PostgreSQL's built-in JSONB, parallelization, and indexing capabilities. This was the most realistic approach to achieve scalability given the client's existing technology stack.

Since the existing system couldn't run "black box" models, Rafael designed a

workflow that translated Scikit-learn ensemble models (Gradient Boosting, Random Forest) into linear rules. This architecture bridged the gap between modern data science and legacy infrastructure. The robust design left behind by Rafael enabled the team to later move from a purely rules-based approach to a full-fledged hybrid solution based on machine learning with very good results.

EXPERTISE

Linux, Linux CLI, PostgreSQL, Python, Pandas, NumPy, Feature Engineering, Fraud prevention, Fraud

04.2024 - 08.2024

Elvia: Energy efficiency: Energy saving

Elvia has established an interdisciplinary committee to support the government's goal of reducing electricity consumption in buildings by 10TWh by 2030 and increasing solar power input by 8TWh. The project, known as ENØK2.0, examines how Elvia can contribute to these goals and assesses the consequences for their grid area, with a focus on the plus customer segment. Data platform representatives worked with the business team to develop data visualizations and proof-of-concepts, assess data quality, and define parameters for plus customers. The project concluded with a review of findings and analyses with the energy coordinators to promote a more data-driven decision-making process.

08.2023 - 08.2024

Elvia: Jordfeil

Elvia must comply with the requirements for monitoring and handling earth faults in accordance with FEF2006 §5-2.

The project will create an automated solution to handle ground faults detected by phase-earth voltage measurements from AMS (Advanced Measurement and Control Systems) in substations and at end customers using leakage current measurements.

The project objectives include:

- Use leakage currents to detect earth faults, and place responsibility for rectification on the right party, with minimal manual follow-up. The expected distribution is 90% end customer and 10% grid company.
- Implement a solution where earth fault detection based on leakage currents generates requirements for customers to correct faults.
- Establish efficient processes for handling ground faults, including changes to substations and the establishment of new substations.
- Plan and carry out necessary retrofits in substations.
- Reduce costs associated with troubleshooting and remediation through safe detection, automation and correct allocation of responsibilities.

ROLLE: Data Scientist

As a data scientist in the project, my responsibilities were to select an appropriate machine learning model architecture, design relevant features/information elements based on the available raw data, set up metrics/evaluation metrics to assess the performance of our model, as well as give presentations to the rest of the team explaining data science concepts relevant to the project.

The underlying type of problem in this project was a so-called multi-class classification problem, where the model predicts probabilities for each category (in this case, the categories are: alert the customer, ignore, mislink, etc.) and the sum of the probabilities is equal to 1.

The metrics we chose to assess the performance of our model were precision and recall, and the probability limits were set by maximizing the F-score. The main choice for domain experts and project managers revolves around determining a relative importance between precision and recall that fits the needs of the project.

During the first few months of the project, we used a specific implementation of the KNN algorithm that was intended to be used with time series data. However, we eventually wanted more features/information elements, such as categorical features, that were not compatible with our time series model architecture. We then redesigned our featurs and started using a random forest model architecture, which improved performance according to our assessment metrics.

The models were trained on a dataset specifically curated for this purpose, and it was lab-tested by the domain experts in our team. Subsequent rounds of improvements to our data labeling solution led to misunderstandings that resulted in mislabeled examples in our dataset. To increase the data quality of our training set, a clustering algorithm was used to identify and correct these poorly lablet examples.

ROLE: Python Developer

I helped the developers in the team (experienced . .NET developers) to get started with setting up a Python application that could serve our machine learning models. I didn't have time to develop myself, but I helped solve obstacles related to both Python in general and specific Python libraries.

EXPERTISE

Flask, Python, Machine Learning, Data Science, SQL, Google BigQuery, dbt, Pandas, NumPy

10.2023 - 08.2024

Elvia: Fortelvia

The project is about exploring and developing generative AI functionality that can help increase productivity.
The team has already produced a chatbot that, with the help of large language models, allows users to chat with multiple internal knowledge bases.

ROLE: Advisor

Generative AI Exploration: I tested new generative AI functionality based on the project manager's requests or curiosity. Then I conducted live demos for the team, where I showed what is possible. For example: explore Langchain in combination with Confluence or BigQuery.

Data science concepts: I explained relevant data science concepts to the project team, so that we had a common understanding.

Regular updates: I kept up to date with the latest research and articles related to generative AI. I then summarized and shared these findings with the team during our weekly sync meetings.

Analysis of user data: I analyzed application user data from consenting users. The reports I made were automated and aggregated to safeguard privacy. I used common statistical tools such as histograms, but also leveraged language models (LLMs) for innovative analyses. For example, I created a report on "user intent" by classifying chatbot conversations using an LLM. In addition, I used LLMs to answer yes/no questions about chatbot interactions, e.g.: "Is the conversation about programming?".

EXPERTISE

SQL, Big Query, Python, Data science, Large Language Models, LangChain, Research

09.2022 - 08.2024

Elvia: Helping economists automate calculations

Rafael helps a couple of economists automate their calculations using Python and Jupyter notebooks.

The customer's workflow includes a number of R-codes published each year by NVE, the Norwegian Water Resources Association and the Norwegian Energy Directorate, which are run behind an API. The customer has experience with R Studio and can run the entire program locally.

There is extensive use of Excel files in the workflow; Excel is both the input and output format. Python libraries are used to read and write Excel files.

Jupyter notebooks are easy to use, and git simplifies collaboration. The complexity has increased over time, and the code is eventually moved to a Python package, which makes the Jupyter experience smoother.

ROLE: Data analyst

Putting themselves in the customer's tasks and understanding their needs. Map out possible ways to simplify and automate their calculations.

ROLE: Python Developer

Implement automation and simplification of the customer's tasks using Python code.

EXPERTISE

Python, Pandas, Jupyter Lab, Git, Microsoft Excel, R

05.2022 - 05.2022

Elvia: Anonymization of customer surveys

Anonymize data from customer surveys and publish it as a data product in the company's internal data catalog.

ROLLE: Data Engineer

I worked with several Excel spreadsheets from a previous project, which contained valuable information from customer surveys. After merging them with other metadata and metrics, it resulted in a number of anonymized data products. The data models were set up in line with the data mesh framework.

EXPERTISE

dbt, Google BigQuery, Python, Pandas, Data Mesh

04.2022 - 09.2023

Elvia: Pegasus

The project explores whether the company's advice and information to customers affect the end customers' electricity consumption

ROLLE: Data Scientist

I worked with the mathematical derivation of 2 metrics, or scores, that can be used to measure 1) the flatness of a spending pattern over time and 2) how well a customer leverages intraday price variations.

I also suggested a possible strategy that uses the 2 metrics mentioned, as well as a consumption prediction solution, to identify scenarios with high potential for improvement: savings for the customer and a reduction in load for the company's infrastructure.

EXPERTISE

dbt, Dataanalysis, Data Mining, Google BigQuery, Big Query, Data science

10.2021 - 04.2022

Elvia: New tariffs

The goal is to help the customer obtain the necessary data extraction they need to update the tariffs, which is done once a year. 2022 is more demanding than usual due to new laws and regulations that introduce a new model for how end customers should be classified MTP. e.g. which effect group they fall into.

The extracts are usually time series where energy consumption / maximum power / number of customers are aggregated across different customer groups. Each customer will be classified based on their 1-year rolling energy consumption, which complicates the calculations and the connections/lineage between the different SQL queries.

The task would have been almost impossible to perform without DBT and BigQuery due to the large amounts of data involved.

In this process, we found errors in measurement values, which are an important basis for the extractions; Work is now being done to correct the errors.

EXPERTISE

dbt, Google BigQuery, Big Data, Database mining, Python

01.2022 -

Elvia: Analyzing electricity price vs. electricity consumption

Efforts are being made to help the customer gain better insight into the relationship between the electricity price and the customers' electricity consumption. This is of great interest to the customer, and is similar to research work.

The work begins by retrieving meter values for historical consumption, as well as historical electricity prices. Then you try to find patterns and correlations. It is also desired to link temperature data into the analysis.

A challenge in this initiative has been that there is a lack of an API that can deliver historical electricity prices in Norwegian kroner, but this is something the customer is working on setting up.

EXPERTISE

Google BigQuery, SQL, Pandas, Python, matplotlib

07.2021 - 08.2024

Elvia: Consumption Prediction / Predict@Scale

This project aims to generate predictions for future energy consumption. In the beginning, the scope was to predict the power consumption of substations (substations). This generated more interest in the project, which was then expanded to also support end-customer-level predictions.

The work consists, among other things, of selecting an appropriate set of so-called features, which can represent the underlying study case in a reasonable way, also known as feature engineering. In addition, we focus on optimizing the use of large amounts of data and minimizing the overall prediction time.

The facts/information elements used were based on: temperature from weather forecasts, electricity price and previous consumption. The time of day, day of the week, and month were also fed into the model as information elements, following a trigonometric transformation.

In terms of the machine learning model architecture used, random forest performed well for substations, where noise levels are relatively low. Predicting future consumption for individual customers is more challenging, due to much greater uncertainties. This led us to adopt probability-based models, which return a probability distribution instead of a single value.

Different probability-based models were tested, and the one that gave us the

best results is the Temporal Fusion Transformer (TFT), which is based on the concept of attention, the main concept driving the current generative AI era. TFT gave us much better results for time series predictions, both at the substation level and at the end customer level.

One of the most important metrics we used to assess the model's performance is quantile coverage. It essentially tells us how well the predicted probability distributions match the expected probability levels.

The project is a continuation of previous work, has a strong focus on automation and on enabling/facilitating access to machine learning models internally within the company, and aims to adopt an ML-Ops approach.

ROLE: Frontend Developer

I helped develop a user interface that enables visualization of time series (consumption development, predictions and prediction errors) and triggers training of new machine learning models.

Training models involves the use of cloud resources, which has costs (such as the creation of compute instances). Therefore, it was important to implement authentication with Open ID Connect. Users sign in to the solution using various authenticated sign-in methods.

ROLE: Backend Developer

I created a Python application using Flask, complete with endpoints to run predictions and train new ML models, among other functionality. In addition, the application handles scheduled prediction runs, writes results to BigQuery for continuous monitoring of prediction errors and real MLOps. It also manages queue messages in Azure Service Bus, breaking down the training process into smaller tasks.

The application's endpoints are secured using the customer's implementation of Open ID Connect. The various instances of the application run as pods in Kubernetes, and the infrastructure is managed with Terraform.

In terms of CI/CD, the pipeline includes Azure DevOps and Aquasec. The documentation is also considered part of the construction process; it is automatically generated with Sphinx and published to GitHub Pages.

ROLLE: Data Scientist

I designed and implemented an ML pipeline that retrieves data, pre-processes it into features expected by the ML model, trains a new ML model, and deploys it.

The ML pipeline runs on Azure AI, and the trained models are deployed there using Azure Container Instances. To ensure consistent data preprocessing and computation of features both during model training and inference, the code is packaged in a separate Python package that is imported by both the ML pipeline and the backend application.

The ML pipeline is scripted in its entirety and is programmatically deployed from the backend application using the Azure AI SDK.

ROLLE: Data Engineer

The raw data, which includes metrics and metadata about both customers and infrastructure, is distributed across different tables and datasets in BigQuery.

This data undergoes a series of transformations (from raw data to curated data and data products) until the desired tables and views are achieved, in line with the principles of the data mesh framework. The entire process is orchestrated using dbt, which also conveniently shows the "lineage" of the data.

EXPERTISE

OpenID Connect, Flask, Google BigQuery, Microsoft Azure ML, HTML, Python, Docker, Microsoft Azure ML Studio, Microsoft Azure DevOps, SQL, Microsoft Azure Kubernetes Service, Maskinlæring, Tidsserier, HashiCorp Terraform, Microsoft Azure Service Bus, dbt, Pandas, Gen AI, Attention-based models

02.2021 - 09.2021

Elvia: POC Facial Recognition

The company has an image system with millions of images, many of which are taken from helicopters or drones. In the context of GDPR regulations, it is desirable to run a facial recognition POC. The goal is initially to detect faces in images, map the scope of personal data in the company's image system, and possibly use bounding-box information from ML predictions to automate deletion/adaptation of the images, e.g. blurring relevant image areas.

ROLE: Backend Developer

The consultant developed a scalable backend application using Flask, which orchestrates the training and exposes the ML models to inference.

The backend application is part of a larger ecosystem, the company's image system, which consists of several other APIs and databases that contain image metadata. The backend application runs on Azure Kubernetes Service, the CI/CD pipeline runs on Azure DevOps, and the infrastructure is orchestrated with Terraform. The API is secured using the company's implementation of OpenID Connect.

The backend application is called by the customer's image system when, for example, new images are added, and the prediction results are stored as image metadata. End users can then filter images in the imaging system based on prediction results.

The API was documented using Sphinx.

ROLLE: Data Scientist

In this role, my main task was to find an image recognition model that could

meet the project goals.

Jupyter notebooks were used extensively for exploratory analysis and image labeling.

A first round of testing was done using standard facial recognition models from cloud services such as Azure Face from Azure Cognitive Services. Azure Face offers various object detection models, which are intended to be used to recognize faces in everyday photos. A challenge in this project, however, was that many of the company's images are taken from helicopters or drones, which is a different area of use than what Azure Face is intended for. This resulted in a high number of false positives, which meant that the Azure Face models were not suitable for the type of images we wanted to do detection on.

In a second round of testing and investigation, we decided to train our own object detection model, using an already trained model (Azure Face) to help us find images with faces to build our training dataset. The prevalence of images with faces was quite low, which meant that we had to scan a large number of production images to find candidates for our training set. Eventually, we built a training kit with around 300 images, and we confirmed that the rate of false positives had dropped significantly, but it was still unacceptably high.

Each dataset iteration was versioned and checked into the codebase. We were particularly concerned with reproducibility and being able to trace a prediction back to the training set. Each prediction was linked to both the dataset version and the application version.

We concluded that the performance of our custom object detection model would continue to improve if trained on multiple images/bounding boxes, but at that point, other tasks were given higher priority and the POC was paused.

EXPERTISE

Microsoft Azure Cognitive Services, Python, Jupyter Notebook, CI/CD, Microsoft Azure DevOps, Sphinx, Flask, Microsoft Azure Kubernetes Service, Object detection, Atlassian Confluence, Atlassian JIRA, Miro boards, SQLite, Microsoft Azure Face API

04.2020 - 03.2021

Elvia: POC Image Recognition

The company's image system contains millions of images, many of which are regularly taken from helicopters or drones. These images contain valuable information about the state of the power grid.

We conducted a proof-of-concept for image recognition, where we:

1. Trains and operates a machine learning model for classification that can classify images based on the type of electric pole shown in the image: wood, steel, or concrete. Later, a fourth category was added: more bars.
2. Trains and operates an object detection machine learning model that detects

if a protective top hood is missing from the top of wood masts.

A Flask back-end application was set up to orchestrate training and to operate the machine learning models. The backend application is called by the customer's image system when new images arrive.

When a new image arrives in the image system, the classification model is called, and the result is stored as image metadata. If the pole is classified as a wooden mast, the object detection model runs, and its result is also stored as metadata. Users can then filter images in the image system by prediction results.

ROLE: Backend Developer

The backend application was written in Python using Flask. Machine learning model training runs on Azure Custom Vision, programmatically triggered from the back-end application via the Azure Custom Vision SDK. Azure Custom Vision is a cloud-based image recognition solution (classification and object detection issues) that conveniently accepts image URLs for both training and inference.

The backend application is part of a larger ecosystem, the company's image system, which consists of several other APIs and databases that contain image metadata. The back-end application runs on Azure Kubernetes Service, the CI/CD pipeline runs on Azure DevOps, and the infrastructure is orchestrated with Terraform. The API is secured using the company's implementation of OpenID Connect.

We used Pytest to write unit and integration tests. The API was documented using Sphinx.

ROLLE: Data Scientist

Jupyter notebooks were used extensively for exploratory analysis and image labeling. We built the training datasets for both models from scratch, a time-consuming process that involved manually tagging many images. The final datasets had about 4500 images for the classification case and 1300 for the object recognition case.

Building the object recognition dataset was a process where we routinely revised and adjusted the rules we set for ourselves regarding how to label the bounding boxes. In the beginning, we only had 2 categories for object recognition: "top cap" and "missing top cap"; However, over time, we increased the number of categories to take into account different aspects such as color ("white top cap", "black top cap") or geometry ("missing top cap", "partially missing top cap").

Each iteration of the datasets was versioned and checked into the codebase. We were particularly concerned with reproducibility and being able to trace a prediction back to the training dataset. Each prediction was linked to both the dataset version and the application version.

The main metric used to assess the model's performance was the F1 score, which we maximized. It reached a whopping 94% for object recognition and 99% for classification. In the object detection case, we conducted several rounds of model evaluation, where we aggregated the categories in different ways, looking for the highest possible F1 score.

A key choice in the object detection case was the model architecture (called "domain" in Azure Custom Vision), where the larger "General A1" model architecture performed much better than "General".

EXPERTISE

Python, Microsoft Azure Cognitive Services, Dataset creation, Image classification, Object detection, Convolutional Neural Network (CNN), Sphinx, Pytest, Jupyter Notebook, Flask, REST, DevOps, CI/CD, Microsoft Azure Kubernetes Service, Azure Cognitive Services, Azure Custom Vision API, Miro boards, Atlassian Confluence, Atlassian JIRA, JWT

03.2020 - 04.2020

Moldtelecom: Moldtelecom Churn POC

We ran a POC for Moldtelecom in collaboration with/on behalf of the Computas Development Center in Romania. The goal was to predict which customers were most likely to cancel their mobile subscription, using an ML model trained with past customer data. In addition, it was desirable to uncover which features were most important, i.e. which variables had the greatest impact.

ROLLE: Data Engineer/Scientist

Pre-processing of the raw data, EDA, training of ML models with different architectures and frameworks, assessment of the models' performance

EXPERTISE

Keras, Jupyter Notebook, Python, Machine Learning, Pandas, Pytorch

09.2019 - 02.2020

Aker Solutions: SubSense

Subsense is a condition monitoring system for an oil and gas subsea platform. This system connects raw data from subsea sensors and subsea services with the use of data analysis based on time series data from IoT (instruments, sensors and various subsea devices). Subsense consists of several analytical applications e.g. design leakage, pump efficiency, choke and valve accumulator, reservoir, hydraulic fluid consumption and instrument condition monitoring.

The different analytical applications deliver different types of results, such as events and time-series data that the domain experts or system monitors can use to more easily see where there could potentially be a problem or which components should be replaced.

ROLLE: Data Scientist

In this role, Rafael worked on designing a general-purpose model that can

handle transformations of different types of sensor data in a flexible and robust way.

ROLLE: Data Engineer

Worked with the transfer and transformation of data between the customer's platform (Cognite) and Google Cloud Platform. The web services developed ran on Google Cloud AppEngine and used Google Cloud Datastore as a database solution.

ROLE: Backend Developer

Rafael worked on implementing the general model as an engine/framework that divides the work into small tasks. The code was written in Python and the tasks run in the cloud using Google AppEngine.
In addition, Rafael helped create and maintain endpoints for the various web services.

EXPERTISE

Python, Google Cloud Platform, Google App Engine, Google Cloud Datastore, Numpy, Pipenv, Flask, Jenkins, Cognite, Atlassian Confluence, Git, Google Cloud Stackdriver, Google Cloud Scheduler, Pytest, Atlassian JIRA, Atlassian BitBucket, Google Cloud Tasks, scipy, IoT, Domenekunnskap: Subsea, Dataanalyse, Pandas

02.2019 - 08.2019

The Government Pension Fund Global: Migration of systems to the cloud

The goal of the project was to migrate different systems from "on-premises" to EC2 instances in AWS. The pipeline involved the use of TeamCity, Octopus Deploy, AWS elastic load balancers, AWS auto-scaling groups and scripting with Windows Powershell to automate the installation of the various software when the instances start up.

ROLE: Developer

Responsible for facilitating solutions into AWS through scripting, testing and deploying them.

EXPERTISE

Octopus Deploy, Windows Powershell, TeamCity, AWS, IAM, Atlassian Confluence, Atlassian JIRA, GitHub, Testrail, Amazon Elastic load, Migrering, Cloud, Skyteknologi, Amazon EC2, Lift&Shift

09.2017 - 08.2019

The Oil Fund: Design, development and system responsibility for a SharePoint-based solution

Design, development, testing and system administrator for an internal personal trading solution: SharePoint 2013 + Nintex Workflows 2013 + Nintex Forms 2013 + JavaScript. The system runs SQL queries against different databases using web services, which involved a lot of database work.

ROLLE: Arkitekt

Define the process and technical requirements together with the customer.

ROLE: Developer

Develop the Sharepoint solution from scratch and put them into production.

ROLE: Test and quality manager

Test the solution and make sure it met the business user's requirements.

ROLE: System administrator

After the solution went live, I became a system administrator. The job was to ensure good operations and correct errors.

EXPERTISE

SharePoint 2013, Microsoft SharePoint Designer 2013, JavaScript, HTML, Nintex Forms, Nintex Workflows, WebServices, System Design, Functional Manager, Requirements Specification, Test, Database, Process Modeling, SQL

09.2016 - 08.2019

The Oil Fund: Parser .NET solution

The web service receives content from different sources and in different formats, with the goal of retrieving a number of desired values. The service converts the content to text and then uses regular expressions to find the desired values. Users communicate with the service through the web app. The app data is stored in a SQL database.
The consultant has worked with the design, development and then maintenance of a .NET solution consisting of an MVC web application and a web service in the back-end.

ROLE: .NET Developer

Design, develop, implement and maintain the business logic of the web service and contribute to the development of the corresponding front-end web application.

ROLE: Integration Developer

Developed the system integration with databases and other software, via web services.

ROLE: Test and quality manager

Design and developed integration tests, general tests and set up relevant KPIs.

EXPERTISE

SQL Developer, HTML, Microsoft .NET, Stored Procedures, C#, Optical Character Recognition, OCR, MVC, SQL, Test, Quality Assurance, Integration Testing, WebServices

2015 - 2016

Hobby project: Investor Playground

The consultant has designed and developed an Android app in his spare time. The app pulls price data from Yahoo! Finance for companies listed on the Oslo Stock Exchange and gives users the ability to design simple trading strategies and see their evolution over time. The app was written in Java using Android Studio

ROLE: JavaDeveloper

The consultant is responsible for the entire process from idea, design, development and implementation of the app.

ROLE: Android Developer

The consultant designed and developed the solution, - App for Android

ROLE: Data analyst

In the role of data analyst, the consultant has modelled, obtained and modelled the analysis of the data.

EXPERTISE

Java, Google Analytics, Android Studio/IntelliJ, Android app design

Education

DURATION	SCHOOL	DEGREE
2003 - 2006	Cuyo Universitet, Argentina	Chemical Process Technology
2006 - 2009	Balseiro Institutt, Argentina	Engineer in nuclear power
2010 - 2014	Norwegian University of Science and Technology (NTNU)	Ph.D. in Chemical Process Engineering

Certifications

DATE	CERTIFICATION	ORGANIZER
01.2020	Google Certified Professional - Data Engineer	Glasspaper

Course

DATA	COURSE	ORGANIZER
11.2023	ML with Berlin 2023	ML Conference
07.2022	EuroPhython (Dublin)	EuroPython Society
09.2021	Norwegian Developers Conference (NDC)	NDC
06.2018	Norwegian Developers Conference (NDC)	NDC
09.2018	Achitecting on AWS	AWS Training and Certification
02.2018	Nordic Infrastructure Conference	NOTHING

Publications

DATA	PUBLICATION
08.2009	Configuration analysis for the RA-6 reactor
12.2014	Modelling and simulation of reactive gas-solid flows in circulating fluidized bed reactors
08.2014	Modeling and simulation of circulating fluidized bed reactors applied to a carbonation/calcination loop
05.2014	Modeling the Chemical Looping Reforming Process Operated in a Circulating Fluidized Bed Reactor Consisting of Two Bubbling Bed Units: Model Validation
12.2016	Numerical modelling and simulation of hydrogen production via four different chemical reforming processes: Process performance and energy requirements

Presentation

DATA	PRESENTATION
12.2012	Lecture at CFD 2012 in Australia

Languages

LANGUAGES	LEVEL
Written and spoken English	Floating
Written and oral Norwegian	Floating
Spanish written and spoken	Native speakers